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# PROJECT AND TEAM INFORMATION

## Project Title

**Smart Traffic and Emergency Route Management**
Dispatcher dashboard for emergency-vehicle routing and traffic-signal preemption

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|---|---|
| *Team Name:*<br>*Team #T079* | *Mentor: Dr. Ashwini Kumar Singh* |
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# PHASE-II PROJECT PROGRESS DESCRIPTION

## Project Abstract

Smart Traffic and Emergency Route Management aims to support faster emergency response by coordinating emergency-vehicle dispatch, route visibility, traffic-signal preemption and driver awareness. Conventional fixed-time signals do not prioritize an approaching ambulance, fire tender or police vehicle, so critical minutes can be lost in congested corridors.

Phase I defined the problem, objectives, proposed architecture and planned modules. In Phase II, the team built a browser-based interactive prototype titled *"Pulse · Emergency Transit & Preemption"*. It brings city operations, incidents, map/corridor visualization, the emergency fleet, signal controls and analytics into one dispatcher-style workspace.

The current deliverable is a front-end prototype (HTML, CSS, JavaScript) with simulation-oriented behaviour. Database connectivity, live traffic/GPS feeds and real signal controllers are treated as planned integrations, not as completed features. The prototype is the foundation for the next stage: persistent data, route calculation and measured response-time evaluation.

## Updated Project Approach and Architecture

**Approach.** Phase II follows a dashboard-first strategy: build the dispatcher interface and simulate emergency workflows before wiring in external services. The workflow is: an incident appears with type, location and assigned unit; the dispatcher selects a unit and opens the map/corridor view; the interface offers signal-priority actions (*Lock Green Wave*, *Override Green*); the activity and analytics views record and summarize the operation.
**Architecture.** The prototype is a single-page web application (`dashboard/index.html`) with embedded CSS and JavaScript and four local vehicle images. Its modules are:

- **Operations dashboard** – incidents, fleet readiness, recent activity.
- **Map/corridor view** – traffic-flow layers, signal nodes, preemption corridor, replay/simulation.
- **Dispatch controls** – Dispatch Next Unit, New Incident, mission actions.
- **Signal controls** – six intersections with Configure Cycle / Override Green.
- **Fleet and analytics** – unit cards, response-time and corridor views, export options.

**Technologies.** HTML, CSS, JavaScript (DOM event handlers, `requestAnimationFrame` simulation loop, `localStorage`). No network protocol is active yet; planned back-end options are Firebase or MongoDB, Google Maps Platform or OSRM, and Dijkstra/A* routing over a road graph.

## Tasks Completed

| Task Completed | Team Member |
|---|---|
| Overall coordination, dashboard architecture and module structure (Overview, Map, Operations, Vehicles, Signals, Analytics); packaging for demonstration | Nishil Bhimani |
| Research on route optimization (Dijkstra / A* direction) and mapping of route-corridor concepts to the map view | Aastik Gupta |
| Dashboard layout, incident cards, fleet cards (ambulance, fire, police, bike medic), signal-intersection cards and data-entity definitions | Jasmin Mangukiya |
| Interactive behaviour: action menus, map/simulation controls, analytics views; source-level testing checklist and demo preparation | Rishi Agarwal |

## Challenges/Roadblocks

- **No live data or backend yet.** The prototype is static front-end code; no database, traffic API or GPS feed is connected. *Plan:* define schemas for vehicles, incidents, routes, signals and events, then connect Firebase or MongoDB for one end-to-end flow.
- **Route engine not integrated.** Route research exists but a tested algorithm is not in the package. *Plan:* implement Dijkstra/A* on a simulated road graph with congestion-aware edge weights.
- **Simulated vs. real data.** Labels such as “Live” and “Realtime” could be mistaken for real measurements. *Plan:* clearly mark simulated data in the UI.
- **Safe signal preemption.** Real controllers need authorization, conflict checks and fail-safe fallback. *Plan:* keep preemption simulated and add timeout and conflicting-approach checks first.
- **Validating analytics.** Response-time charts lack a defined dataset. *Plan:* define baseline, sample size and calculation method before reporting results.

## Tasks Pending

| Task Pending | Team Member (to complete the task) |
|---|---|
| Database layer: schemas, persistence of incidents, units and operation logs | Jasmin Mangukiya |
| Route engine (Dijkstra/A*) with congestion-aware weights and map/routing integration | Aastik Gupta |
| Dispatch and signal-control APIs with priority logic, authentication and fail-safe handling | Nishil Bhimani |
| Response-time experiments, browser/export testing and final demo | Rishi Agarwal |

# Project Outcome/Deliverables

- A dispatcher-style web dashboard covering city operations, map/corridor view, operations dispatch, emergency fleet, traffic signals and analytics.
- Simulation of emergency dispatch and signal-priority actions (green-wave lock, green override).
- A database design for vehicles, incidents, routes, signals, operation events and response metrics.
- A congestion-aware route-optimization module (Dijkstra/A*) on a road graph.
- Measured comparison of response times with and without preemption, and the final report and demo.

# Progress Overview

The interface layer of the project is largely complete: the dashboard has dedicated views for operations, map, dispatch, fleet, signals and analytics, with incident-specific actions and simulation controls. This is the strongest Phase II achievement and is on schedule.

The back-end side is behind the plan: no database connection, live traffic/GPS data, route algorithm integration or measured response-time dataset is in place yet. Nothing is ahead of schedule.

For the next phase the team will prioritize one small, testable end-to-end flow – incident creation, unit assignment, route calculation, event persistence and response-time comparison – before adding more interface features. Signal preemption will remain simulated and safety-checked.

# Codebase Information

**Repository:** https://github.com/nishilbhimani/Smart-Traffic-Emergency-Route-Management-.git
**Package structure:** `dashboard/index.html` (HTML, embedded CSS and JavaScript) with the local images `ambulance.jpg`, `fire_tender.jpg`, `bike_medic.jpg` and `police_interceptor.jpg`.

# Testing and Validation Status

| Test Type | Status (Pass/Fail) | Notes |
|---|---|---|
| Source and package inspection | Pass | Page structure, navigation, controls and four image assets present |
| Navigation between main tabs | Pass (source); live check pending | Overview, Map, Operations, Vehicles, Signals, Analytics |
| Incident action menus | Pass (source); live check pending | Context-specific actions found in source |
| Map and simulation controls | Pass (source); live check pending | Replay, speed and layer controls |
| Fleet and signal views | Pass (source); live check pending | Unit cards with images; six intersection cards |
| Analytics and export | Pending | Export output to be verified |
| Database persistence, live GPS, real signal | Not tested | Not implemented yet; planned for next phase |

# Deliverables Progress

- Dispatcher dashboard (operations, map, fleet, signals, analytics): **Completed** (prototype).
- Simulated dispatch and signal-priority actions: **Completed** (interface level).
- Database design and persistence: **In progress** (entities defined, not connected).
- Route-optimization module (Dijkstra/A*): **In progress** (research done, not integrated).
- Response-time comparison and analytics validation: **Pending**.
- Final report and demo: **Pending**.